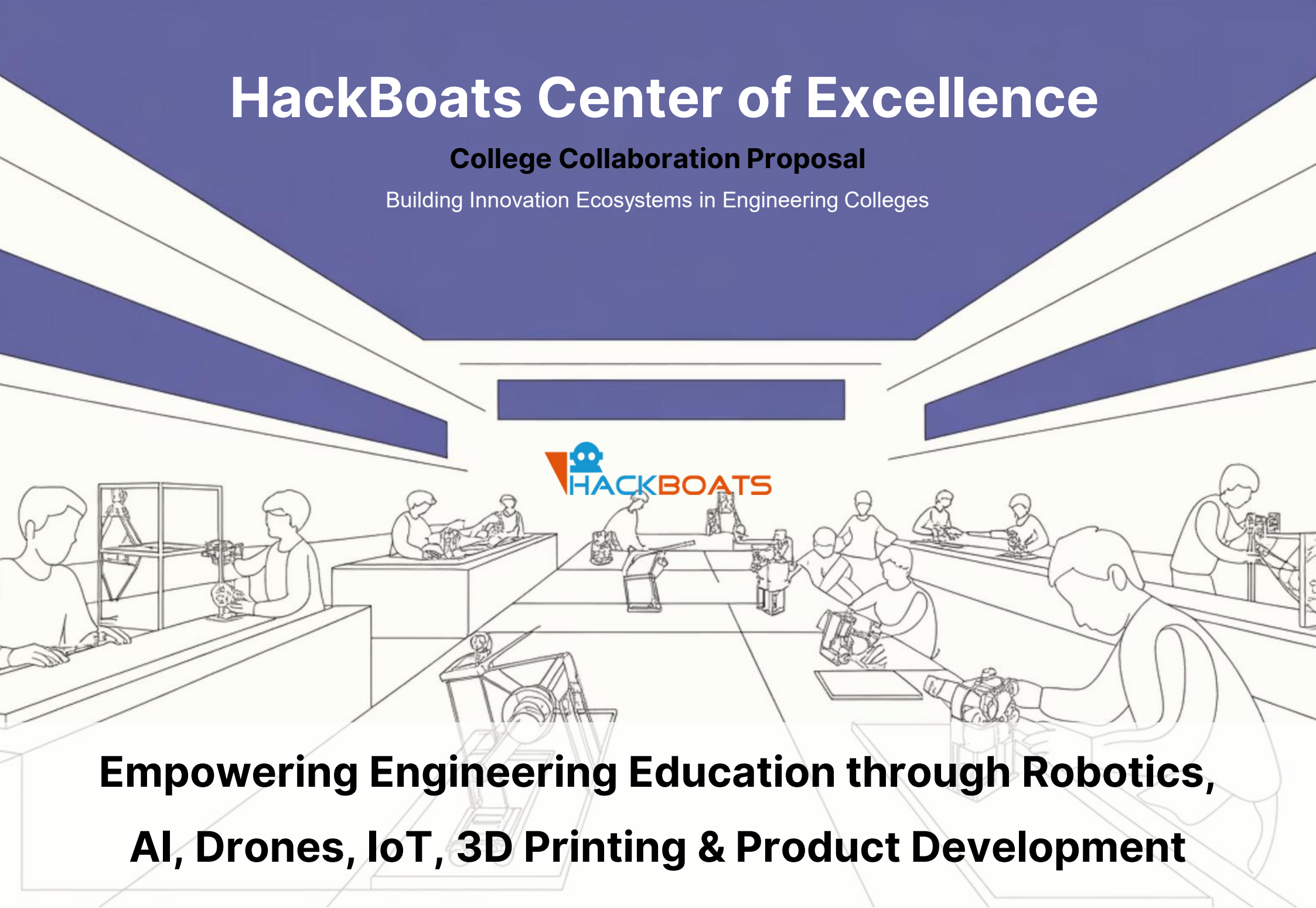
HackBoats Center of Excellence

College Collaboration Proposal

Building Innovation Ecosystems in Engineering Colleges



Empowering Engineering Education through Robotics, AI, Drones, IoT, 3D Printing & Product Development

About HackBoats

HackBoats stands at the forefront of innovation in engineering education, bringing over **12+ years** of specialized expertise in transforming how students learn and apply cutting-edge technologies. Our comprehensive approach encompasses **Robotics, Artificial Intelligence, Drone Technology, Internet of Things (IoT), Product Development, and advanced 3D Printing solutions.**

12+

Years of Experience

Proven track record in engineering education transformation

200+

Workshops Conducted

Across multiple engineering colleges in India

27+

College Partners

Building innovation ecosystems nationwide

150K+

Students Trained

Transformed into industry-ready engineers

With a proven track record spanning the length and breadth of India, we've helped thousands of students transition from theoretical knowledge to practical, industry-ready skills. Our experienced team of engineers, innovators, and educators works tirelessly to bridge the gap between academic learning and real-world application, making every engineering student truly competitive in today's technology-driven job market.

> **Our Mission:** "To make every engineering student industry-ready by turning colleges into innovation hubs."

Objective of the Collaboration

- In today's rapidly evolving technological landscape, engineering colleges face the critical challenge of equipping students with industry-relevant skills alongside theoretical knowledge. **HackBoats** addresses this challenge head-on through a comprehensive collaboration model that establishes **a Centre of Excellence** within your institution—at **infrastructure or no capital cost to the college.**

- This initiative is more than just a program; it's a strategic investment in your institution's future. By partnering with HackBoats, your college gains access to an extensive industry network, proven pedagogical frameworks, and continuous expert mentorship—all dedicated to fostering a sustainable culture of innovation and entrepreneurship on campus.

- This groundbreaking collaboration transforms your campus into a vibrant innovation hub, empowering students to explore, experiment, and excel in emerging technologies such as Robotics, AI, IoT, 3D printing, product development and Drones.

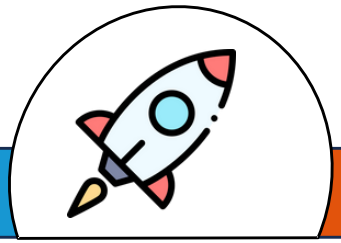
- By joining the HackBoats network of progressive institutions, your college will be positioned as a forward-thinking leader in engineering education, attracting top talent and recognition as a premier hub for technological innovation—all without any financial burden.

Explore
Students discover new technologies through guided exploration and curiosity-driven learning

Experiment
Hands-on projects enable practical application of theoretical concepts in real-world scenarios

Excel
Empowered learners achieve excellence and become innovators ready for industry challenges

Position your college as a leader in engineering education — this partnership delivers a competitive advantage without financial burden, creating lasting value for students, faculty, and the institution's reputation in the academic community.



Objective of the Collaboration

HACKBOATS

Zero-Cost Setup

Complete lab infrastructure sponsored by HackBoats with no upfront investment required from the institution

Innovation Focus

Cutting-edge curriculum designed to foster creativity, problem-solving, and entrepreneurial thinking

Skill Development

Industry-aligned training programs that enhance student employability and career readiness

Enhanced Student Skills

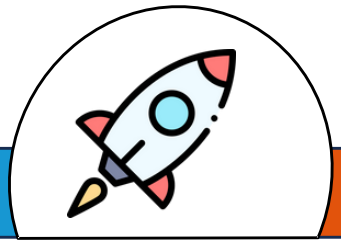
Provide hands-on training in cutting-edge technologies including Robotics, AI, Drones, IoT, and 3D Printing, making students truly industry-ready and competitive in the job market.

Improved Rankings

Significantly boost your college's NAAC, NBA, and NIRF metrics through enhanced innovation indicators, research output, and student achievement benchmarks.

Industry Linkage

Establish stronger connections with technology companies and startups, creating pathways for internships, placements, and collaborative projects that benefit students directly.



Technology Curriculum Overview

Our curriculum is meticulously designed to progress from fundamental concepts to advanced applications, ensuring that students build a strong foundation before tackling complex challenges. Each module is carefully scaffolded, with theoretical understanding immediately reinforced through practical implementation, creating deep, lasting learning that transcends mere memorization.

Fundamentals

Electronics basics, circuit theory, programming logic, and introductory robotics concepts

Application

Microcontrollers, sensor integration, actuator control, and IoT systems development

Innovation

AI implementation, machine learning integration, autonomous systems, and prototype development

Robotics Track

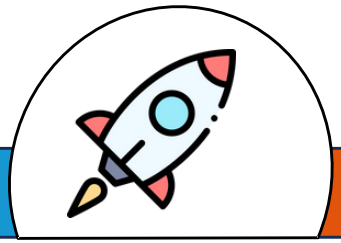
- Mechanical design principles
- Motor control systems
- Path planning algorithms
- Sensor fusion techniques
- Autonomous navigation
- Human-robot interaction

IoT Track

- Embedded systems programming
- Wireless communication protocols
- Cloud connectivity
- Data visualization dashboards
- Edge computing concepts
- Smart device development

AI Track

- Machine learning fundamentals
- Neural network architectures
- Computer vision applications
- Natural language processing
- Reinforcement learning
- Ethical AI considerations



Industry Partnerships & Career Pathways

HackBoats maintains strategic relationships with leading technology companies, innovative startups, and research organizations across India. These partnerships create direct pathways from education to employment, ensuring that students don't just learn skills—they gain access to opportunities that launch meaningful careers in technology and innovation sectors.

Internship Placement

Direct connections to companies seeking skilled interns, with top students receiving priority referrals and personalized mentorship from industry professionals actively involved in our programs.

Campus Recruitment

Partner companies conduct focused recruitment drives specifically for HackBoats program alumni, recognizing the quality and practical orientation of our training methodology.

Entrepreneurship Support

Students with viable startup ideas receive guidance, mentorship, and potential funding connections through our innovation incubation network and angel investor partnerships.

Partner Company Sectors

- Robotics and automation firms
- IoT solution providers
- AI and machine learning companies
- Product development startups
- Manufacturing technology enterprises
- Research and development organizations
- Smart systems integrators

Career Outcomes

- Hardware engineer positions
- Embedded systems developers
- IoT solution architects
- Robotics programmers
- AI/ML engineers
- Technical entrepreneurship
- Innovation consultants

Centre of Excellence Concept

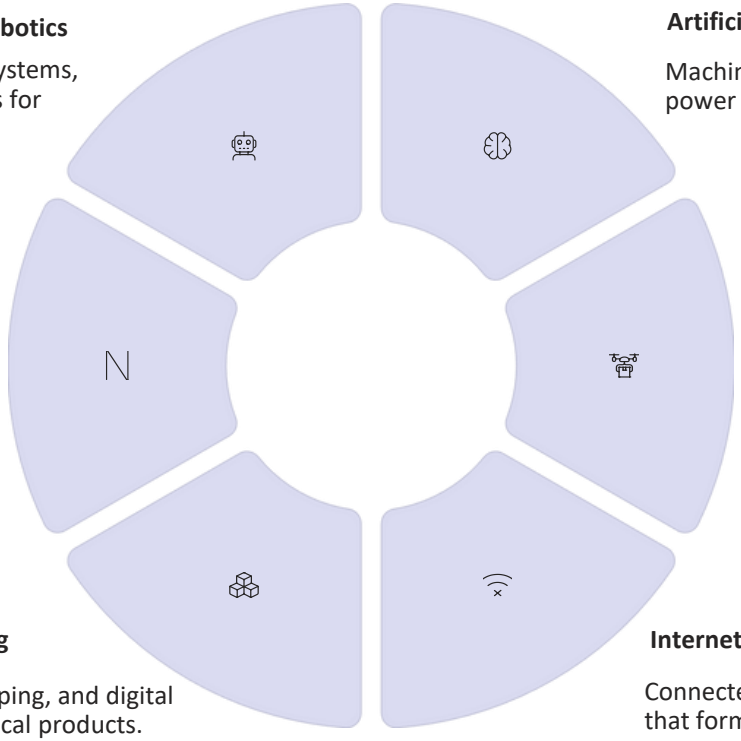
The HackBoats Centre of Excellence is a comprehensive, fully-sponsored innovation ecosystem designed to seamlessly integrate into your college infrastructure. This model represents a paradigm shift in how engineering colleges can offer advanced technical training without capital expenditure or operational overhead.

Robotics

From basic automation to advanced autonomous systems, students learn to design, build, and program robots for real-world applications.

Artificial Intelligence

Machine learning, neural networks, and AI algorithms that power tomorrow's intelligent systems and applications.



Product Development

End-to-end innovation process from ideation to market-ready prototypes using design thinking methodologies.

Drone Technology

Aerial robotics, flight controllers, and autonomous navigation systems for commercial and industrial applications.

3D Printing

Additive manufacturing, rapid prototyping, and digital fabrication for turning ideas into physical products.

Internet of Things

Connected devices, sensor networks, and smart systems that form the backbone of Industry 4.0.

The Sponsored Model

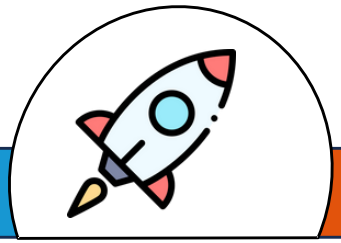
This is a **zero-investment opportunity** for your college. HackBoats provides all equipment, training materials, expert instructors, and ongoing support. Your college simply provides the physical space and coordination assistance.

1 → 2 → 3

College
Provides space & coordination

HackBoats
Provides setup & programs

Students
Gain cutting-edge skills



HackBoats Student Programs

Our comprehensive program portfolio is strategically designed to meet students at every stage of their engineering journey, providing progressively advanced learning experiences that build upon foundational knowledge while introducing new concepts and technologies. Each program is carefully crafted to align with students' academic progression and career development needs.

Engineering Foundation

Target: First-year students | **Investment:** ₹499

A comprehensive introduction to engineering fundamentals covering basic electronics, programming logic, and introductory robotics. Students learn core concepts through hands-on projects, building a strong foundation for advanced learning.

Jugaad Program

Target: Second-year students | **Investment:** ₹499

Celebrating the spirit of ingenious problem-solving, Jugaad focuses on resourceful innovation, prototype development, and entrepreneurial skills. Students tackle real-world challenges with creative, cost-effective solutions.

Career & Innovation Program

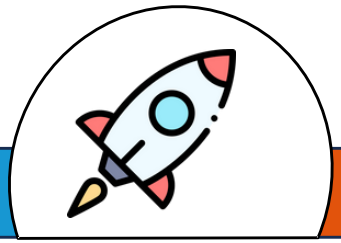
Target: Third-year students | **Investment:** ₹499

Focused on career development and innovative thinking, this program introduces students to advanced robotics, IoT systems, and AI concepts while emphasizing project-based learning and industry applications.

Hackathons

Target: All years | **Investment:** Based on Event type

Regular competitive events where students collaborate intensively to build innovative solutions. These events foster teamwork, rapid prototyping skills, and exposure to industry judges and potential employers.



Free Training for Selected Innovators Inside the Centre of Excellence

At **HackBoats**, we believe every passionate learner deserves a chance to innovate and excel. Whilst we conduct various paid training programmes outside the Centre of Excellence, we recognise exceptional talent and dedication. The best-performing and most committed students from those programmes are selected for exclusive advanced training inside the CoE — completely free of cost.

This unique opportunity ensures that financial constraints never hold back true innovators. Our selection process identifies students who demonstrate outstanding performance, creativity, and commitment during external HackBoats programmes and innovation challenges. Once selected, these talented individuals gain access to world-class facilities, cutting-edge technology, and personalised mentorship that would otherwise be inaccessible.

Merit-Based Selection

Students are chosen based on exceptional performance in external HackBoats programmes and innovation challenges, ensuring only the most dedicated learners advance.

Advanced Hands-On Learning

Selected innovators receive intensive training in Robotics, Artificial Intelligence, Internet of Things, Drone Technology, and 3D Printing.

Expert Mentorship

One-on-one guidance from HackBoats industry experts who bring real-world experience and insights to accelerate learning.

100% Free Training

Whilst students invest in external programmes, all CoE advanced training is provided at absolutely no cost — removing financial barriers to innovation.

Investment Model: Students pay only for external HackBoats training programmes. Once selected, all advaned training inside the Centre of Excellence is provided completely, including access to equipment, mentorship, and facilities.

Benefits for Students

Participation in HackBoats programs transforms students from passive learners into active innovators. The hands-on nature of our training ensures that students don't just understand concepts theoretically—they know how to apply them to solve real problems. This practical experience is invaluable when entering the job market or pursuing higher education.

Industry-Recognized Certifications

Every completed program comes with a professional certification that holds weight with employers and demonstrates concrete skills and project experience to recruiters and admissions committees.

Internship Opportunities

Top performers gain access to our network of partner companies for internships, providing real industry experience and potential full-time employment pathways after graduation.

Portfolio Development

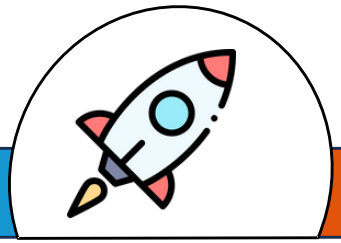
Build an impressive portfolio of completed projects that demonstrate capabilities to potential employers and graduate schools, setting you apart from other candidates in competitive processes.

Innovation Mindset

Beyond technical skills, students develop critical thinking, problem-solving abilities, and the confidence to tackle complex challenges—skills that are valuable in any career path they choose.

Industry Exposure

Regular workshops with industry experts, guest lectures from leading technologists, and exposure to cutting-edge developments ensure students stay current with technological trends and market demands.



Benefits for Colleges

The HackBoats Centre of Excellence offers transformative benefits for partner colleges, positioning them as leaders in engineering education while requiring zero capital investment. This collaboration directly addresses key institutional goals including accreditation improvements, placement enhancement, and reputation building.

Enhanced NAAC/NBA/NIRF Rankings

Innovation indicators are critical metrics for accreditation bodies. The Centre provides documented evidence of innovation and entrepreneurship activities, industry collaboration and partnerships, student project outcomes and certifications, research and development initiatives, and quality enhancement mechanisms that directly improve institutional rankings.

Improved Placement Statistics

Students with hands-on experience in emerging technologies are significantly more employable. Benefits include higher placement rates in technology companies, better salary packages due to specialized skills, increased participation in campus recruitment drives, and alumni success stories and testimonials that attract future students.

Institutional Reputation

Position your college as a forward-thinking, industry-aligned institution with media coverage of innovation events and hackathons, recognition in technology and education circles, increased student applications and enrollment, stronger alumni engagement and pride, and enhanced parent and stakeholder confidence.

Forecasted Impact: With over **12 years of collaboration** experience, we anticipate partner colleges will see up to a 25% boost in quality placements and notable gains in NIRF rankings within two years of launching the Centre of Excellence.

Accreditation & Quality Enhancement

Modern accreditation frameworks increasingly emphasize innovation culture, industry collaboration, and outcomes-based education. The HackBoat partnership directly addresses multiple criteria across NAAC, NBA, and NIRF evaluation parameters, providing documented evidence of institutional commitment to student-centric, experiential learning that extends beyond traditional classroom instruction.

NAAC Criteria Support

Criterion 1.1.3: Institution integrates cross-cutting issues relevant to professional ethics, gender equity, environment, and sustainability into curriculum

Criterion 3.4: Institution provides adequate facilities for physical and academic growth of students

Criterion 5.3: Student participation and activities demonstrating innovative learning

NBA Outcome Alignment

Program Outcome 2: Problem analysis and design/development of solutions

Program Outcome 3: Design/development of solutions meeting specified needs

Program Outcome 5: Use of modern tools for engineering activities

Program Outcome 12: Lifelong learning and independent learning skills

Documentation support includes detailed reports on student enrollment, participation metrics, project outcomes, skill development assessments, and industry engagement activities. These comprehensive records strengthen accreditation applications while simultaneously serving as internal quality improvement data, enabling evidence-based decision-making about program enhancements and institutional strategy refinement.

Zero-Cost Investment Model

No Financial Burden

Unlike traditional lab setups that require substantial capital expenditure, equipment maintenance costs, and ongoing operational expenses, the HackBoats Centre operates on a completely self-sustaining model. The college invests **zero rupees** in equipment, materials, or instructor salaries.

HackBoats handles all financial aspects including:

- Complete lab equipment and technology infrastructure
- Training materials, kits, and consumables
- Expert instructor salaries and benefits
- Program development and curriculum updates
- Marketing and student enrollment activities
- Maintenance and equipment upgrades

01

College Provides Space

Allocate a dedicated room (approximately 500-800 sq ft) for the Centre of Excellence with Required Storage and Furniture

02

Appoint Coordinator

Designate a faculty member to coordinate activities with HackBoats team

03

HackBoats Invests

Complete setup including equipment, materials, and mentor at our expense

04

Programs Launch

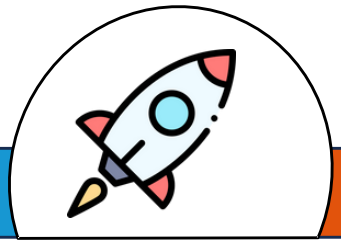
Student enrollment begins with revenue covering all operational costs

05

College Benefits

Consultancy projects Optional 20% revenue share plus all reputational and accreditation benefits

Optional Revenue Sharing: Colleges can choose to participate in a 20% revenue-sharing model, creating an additional income stream from student program enrollments while still maintaining zero initial investment.



Implementation Timeline

The establishment of a HackBoats Centre of Excellence follows a carefully planned, phased approach that ensures smooth integration with your college's existing infrastructure and academic calendar. The entire process, from initial agreement to full operational status, typically takes 8-12 weeks.

1

Week 1-2: MoU Singing and Centre of Excellence inauguration

2

Week 3-4: Centre for Excellence Eligibility Test (CEET)

CEET will be used to select eligible students for induction into the Centre of Excellence.

3

Week 5–6: CoE Induction Program for students

Conducted over a total of 9 days, with separate 3-day sessions for 1st, 2nd, and 3rd year students

4

Week 6-8: Centre of Excellence deployment

5

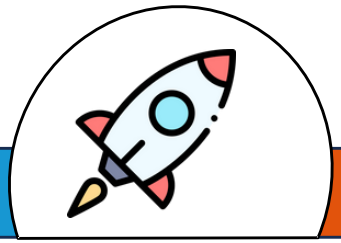
Week 8-10: Program Expansion

Launch of additional year-wise programs based on enrollment demand. Marketing campaigns target specific student cohorts for maximum participation.

6

Week 10-12: Events Launch

First hackathon event planning and execution. Inter-college competitions, exhibitions, and innovation showcases that generate visibility across campus.



Proposed MoU Structure

The Memorandum of Understanding between HackBoats and your institution establishes a clear framework for collaboration, defining mutual responsibilities, expectations, and terms of engagement. This structured agreement ensures transparency, accountability, and alignment of objectives throughout the partnership duration.

Partnership Duration

Initial term of **3 years** with automatic renewal options. . Either party may terminate with 90 days notice after the first year.

Mutual Commitments

Defined responsibilities for both HackBots and the institution

Revenue Sharing

Transparent financial model with optional institutional benefits

HackBoats Contributions

1. Complete lab equipment sponsorship
2. Curriculum development and regular updates
3. Trained instructors and technical mentors
4. Learning materials and project kits
5. Marketing and promotional support for enrollments
6. Certification issuance and industry connections
7. Ongoing maintenance and equipment replacement
8. Performance tracking and reporting systems

Institutional Contributions

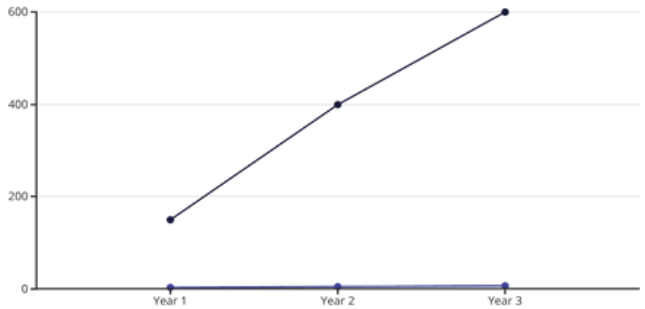
1. Dedicated lab space with basic amenities and furniture along with storage racks
2. Student enrollment facilitation and promotion
3. Faculty coordinator assignment for liaison
4. Scheduling support and timetable integration
5. Access to student communication channels
6. Participation in orientation and awareness events
7. Brand association and co-promotion
8. Feedback mechanisms and quality assurance

Financial Model & Sustainability

Self-Sustaining Operations

The HackBoats Centre operates on a proven financial model that ensures long-term sustainability without requiring ongoing subsidies from the college or external funding sources. Student program fees cover all operational expenses while maintaining affordability for students.

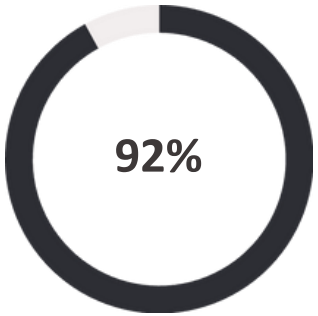
This pricing strategy is carefully calibrated based on years of experience across 27+ colleges. It's affordable enough to ensure broad student participation while generating sufficient revenue to maintain high-quality instruction, modern equipment, and comprehensive materials.



Typical growth pattern shows steady enrollment increases as word-of-mouth spreads and program reputation strengthens within the student community. We forecast that most centres reach optimal enrollment by the second semester.

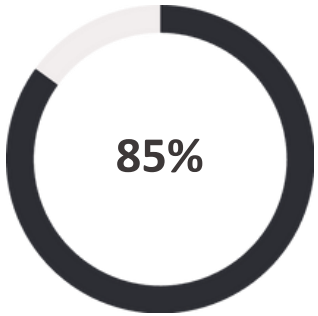
Revenue Streams:

- Student program enrollments (primary revenue)
- Corporate-sponsored hackathons and competitions
- Industry partnership and placement facilitation fees
- Advanced certification programs for seniors
- Faculty development training workshops



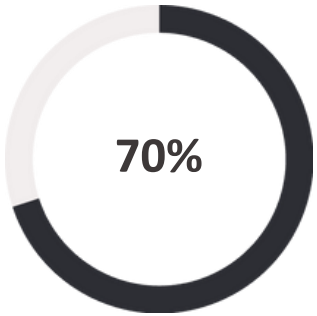
Student Satisfaction Rate

Based on post-program surveys across all partner colleges



Program Completion Rate

High completion rates indicate strong student engagement



Repeat Enrollment Rate

Students returning for advanced programs in subsequent years

Note : All the Statistics which we included above are cumulative reviews from partners colleges for the programs that we are implementing for past 12 years

Long-Term Impact & Success Stories

The true measure of the HackBoats Centre of Excellence lies not just in numbers, but in the transformed lives and careers of students who have participated in our programs. Across our network of 27+ partner colleges, we've witnessed remarkable outcomes that validate our approach to hands-on, practical engineering education.

Career Acceleration

Students who complete HackBoats programs report 40% higher starting salaries compared to peers, with many receiving multiple job offers from top technology companies. Several alumni have joined giants like Amazon, Google, and Microsoft directly after graduation.

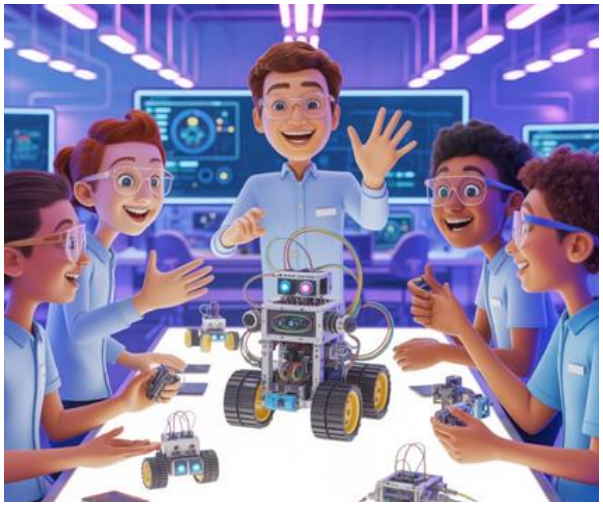
Entrepreneurship Success

Over 50 startups have emerged from projects initiated in HackBoats Centers, with 12 successfully raising seed funding. These student entrepreneurs are creating jobs and contributing to the innovation ecosystem while still in college or shortly after graduation.

Competition Victories

Students from HackBoats partner colleges have won numerous national and international competitions including Smart India Hackathon, IIT Tech Fests, and industry-sponsored innovation challenges, bringing recognition to their institutions.

Why Choose HackBoats ?



Proven Expertise & Track Record

HackBoats stands apart from other education technology providers through our deep understanding of both technology and pedagogy. Our team comprises engineers who have worked in leading technology companies, educators who understand student learning patterns, and innovators who stay ahead of industry trends.

With over **200 workshops conducted** and **27+ successful college partnerships**, we've refined our approach to deliver maximum impact. We understand the unique challenges engineering colleges face and have designed solutions that work within existing constraints while delivering exceptional outcomes.

1

Industry-Aligned Curriculum

Our programs are continuously updated to reflect current industry needs and emerging technology trends, ensuring students learn skills that employers actually want.

2

Expert Instructors

All our trainers have significant industry experience and are passionate educators who know how to make complex concepts accessible and engaging.

3

Comprehensive Support

From initial setup through ongoing operations, our team provides continuous support, quality monitoring, and program improvements based on feedback.

4

Proven Results

Measurable outcomes including improved placements, higher rankings, and enhanced student satisfaction demonstrate the tangible value of our programs.

Contact Information

Get in Touch

We're eager to discuss how the HackBoats Centre of Excellence can be customized for your institution's specific needs and goals. Our team is available to answer questions, provide additional information, and schedule campus visits at your convenience.

Whether you're a Principal looking to enhance your college's reputation, a Dean seeking to improve department outcomes, or a faculty member passionate about student success, we welcome the opportunity to explore this partnership with you.

Website

https://hackboats.com/

Contact number

888-60-999-57

Likhith –Head Operations, Centre of Excellence.

Location

Hyderabad, India

Next Steps

Call Us

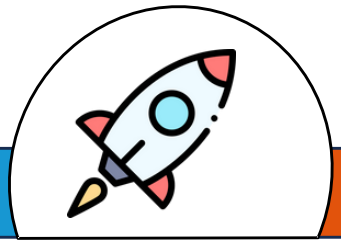
Schedule a consultation call to discuss your specific requirements and timeline

Campus Visit

Arrange for our team to visit your college and present detailed proposal to stakeholders

Partnership Begins

Sign MoU and launch your Centre of Excellence, transforming student futures



Building the Future Together

Let's Create Tomorrow's Innovators Today

The future of engineering education is hands-on, practical, and deeply connected to industry needs. The HackBoats Centre of Excellence brings this future to your campus today—at zero cost to your institution. This is more than a proposal; it's an invitation to join a movement transforming engineering education across India.

27+	150K+	0	100%
Partner Colleges	**Students Trained**	**Investment Required**	**Commitment to Excellence**
Growing innovation network across India	Lives transformed through hands-on learning	Complete zero-cost model for your institution	Quality and innovation in everything we do

Ready to Transform Your Institution?

Contact HackBoats today to schedule a detailed presentation and campus visit. Let's discuss how we can customize the Centre of Excellence for your college's unique needs, student population, and institutional goals. Together, we can build the future of engineering education.

HackBoats Innovations

Empowering the next generation of engineering innovators.

> "Building the Future of Engineering Education."